THREE EQUIVALENT CHART TESTS FOR RELATIVE PSEUDO-COHERENCE OF COMPLEXES ON ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10761-10784 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 10786-10831, the complete author proof (44 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side (both complete). Central glue card for the spaces-side relative-pc toolkit of waves 12-18.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. More on Morphisms dictionary: qcoh-relative-pseudo-coherence-characterize, relative-pseudo-coherent-descends-fppf (L16106), relative-pseudo-coherent-post-compose (L16168), pull-relative-pseudo-coherent, flat-finite-presentation-pseudo-coherent.

Source reference. Original lemma 0695: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10831–10831. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09US: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10831–10831. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09VF: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10831–10831. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CSN: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10831–10831. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CSP: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10831–10831. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CSW: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10746–10759. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is locally of finite type. Let $m \in \mathbf{Z}$. Let $E \in D_{QCoh}(\mathcal{O}_X)$. With notation as explained in Remark 0CSW the following are equivalent:*

(1) *for every commutative diagram*

$$\begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

where U, V are schemes and the vertical arrows are étale, the complex $E|_U$ is m-pseudo-coherent relative to V,

(2) *for some commutative diagram as in (1) with $U \to X$ surjective, the complex $E|_U$ is m-pseudo-coherent relative to V,*

(3) *for every commutative diagram as in (1) with U and V affine the complex $R\Gamma(U, E)$ of $\mathcal{O}_X(U)$-modules is m-pseudo-coherent relative to $\mathcal{O}_Y(V)$.*

Proof. Part (1) implies (3) by More on Morphisms, Lemma 09VF.

Assume (3). Pick any commutative diagram as in (1) with $U \to X$ surjective. Choose an affine open covering $V = \bigcup V_j$ and affine open coverings $(U \to V)^{-1}(V_j) = \bigcup U_{ij}$. By (3) and More on Morphisms, Lemma 09VF we see that $E|_U$ is m-pseudo-coherent relative to V. Thus (3) implies (2).

Assume (2). Choose a commutative diagram

$$\begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

where U, V are schemes, the vertical arrows are étale, the morphism $U \to X$ is surjective, and $E|_U$ is m-pseudo-coherent relative to V. Next, suppose given a second commutative diagram

$$\begin{array}{ccc} U' & \longrightarrow & V' \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

with étale vertical arrows and U', V' schemes. We want to show that $E|_{U'}$ is m-pseudo-coherent relative to V'. The morphism $U'' = U \times_X U' \to U'$ is surjective étale and $U'' \to V'$ factors through $V'' = V' \times_Y V$ which is étale over V'. Hence it suffices to show that $E|_{U''}$ is m-pseudo-coherent relative to V'', see More on Morphisms, Lemmas 0CSN and 0CSP. Using the second lemma once more it suffices to show that $E|_{U''}$ is m-pseudo-coherent relative to V. This is true by More on Morphisms, Lemma 09US and the fact that an étale morphism of schemes is pseudo-coherent by More on Morphisms, Lemma 0695. □